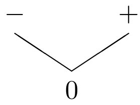
(a) Axiomatics

The collections of sign vectors that make up an oriented matroid are highly structured, and not just random collections. In fact, they arise in the following way.

For any linear subspace $U \subseteq \mathbb{R}^n$, we define the operator SIGN as

$$\mathrm{SIGN}(U) \quad := \quad \{\mathrm{sign}(\boldsymbol{x}) : \boldsymbol{x} \in U\} \quad \subseteq \quad \{+,-,0\}^n.$$

There is a natural *partial order on the set of signs* $\{+,-,0\}$: we set $0 < +$ and $0 < -$, while $+$ and $-$ are incomparable:



This corresponds to the fact that a number that is slightly perturbed either keeps its sign, unless it is zero, in which case its sign can change to $+$, or to $-$, or remain 0.

On sets of sign vectors $\mathcal{S} \subseteq \{+,-,0\}^n$, we use componentwise partial ordering: $\text{u} \leq \text{u}'$ if and only if $u_i \leq u_i'$ holds for all positions i. Thus we get, for example,

$$(0{+}0{+}000{-}{-}{+}{-}) \quad < \quad (0{+}{-}{+}{-}{+}0{-}{-}{+}{-})$$

but

$$(0{+}0{+}000{-}{-}{+}{-}) \quad \not\leq \quad (00{-}{+}{-}{+}0{-}{-}{+}{-})$$

because of the second position, where $+ \not\leq 0$.

We use the componentwise partial order to define the operator MIN, which takes all the *minimal nonzero sign vectors* in $\mathcal{S}$:

$$\mathrm{MIN}(\mathcal{S}) \quad := \quad \{\text{u} \in \mathcal{S}\backslash\mathbf{0} :\ \text{there is no } \text{u}' < \text{u} \text{ with } \text{u}' \in \mathcal{S}\backslash\mathbf{0}\}.$$

We will apply the operator SIGN equally to sets of row vectors and of column vectors. Similarly, we apply MIN both to sets of row sign vectors and of column sign vectors.

With these conventions, we can describe the oriented matroid data for V as follows:

$$\mathcal{V}(V) = \mathrm{SIGN}(\mathrm{Dep}(V)) \quad \mathcal{C}(V) = \mathrm{MIN}(\mathrm{SIGN}(\mathrm{Dep}(V))) = \mathrm{MIN}(\mathcal{V}(V))$$
$$\mathcal{V}^*(V) = \mathrm{SIGN}(\mathrm{Val}(V)) \quad \mathcal{C}^*(V) = \mathrm{MIN}(\mathrm{SIGN}(\mathrm{Val}(V))) = \mathrm{MIN}(\mathcal{V}^*(V))$$

Furthermore, the spaces $\mathrm{Dep}(V)$ and $\mathrm{Val}(V)$ determine each other by Proposition 6.4, so all four sets of data are determined by $U := \mathrm{Val}(V)$. In this sense, we talk about "the oriented matroid $\mathcal{M} = \mathcal{M}(U)$ of the subspace $U \subseteq \mathbb{R}^n$." The dimension $r := \dim(U)$ is called the *rank* of $\mathcal{M}$. So what we were studying in Sections 6.1 and 6.2 were the vectors and circuits of two realizable oriented matroids of rank r respectively $n-r$,

$$\mathcal{M} = \mathcal{M}(\mathrm{Val}(X)) \qquad \text{and} \qquad \mathcal{M}^* = \mathcal{M}(\mathrm{Dep}(X)).$$

It is easy to write down extensive lists of axioms that are satisfied by any collection $\mathrm{SIGN}(U)$. So one gets to *axiom systems* for oriented matroids. We'll get to that in Lecture 7, but without much detail. In fact, the proofs relating various axiom systems for oriented matroids tend to involve hard work, something we try to avoid on this show. (This is, however, at the basis of oriented matroid theory; we refer to [96, Ch. 3]).

The oriented matroid of a point configuration is a delicate model for its geometry. It provides a much finer model than what the *matroid* encodes about a vector configuration. (If you want to know what a matroid is, see Welsh [555], White [557], or Oxley [430].) One can prove that in fact the approximation of the combinatorial model to "geometric reality" is extremely good — this is made precise in Lawrence's "topological representation theorem." We'll get back to this in Lecture 7.

This means also that all the main features of the geometry of vector configurations can be derived from formal properties (the axioms). In fact there are very few geometric statements that would be true for vector configurations but fail for oriented matroids — so whatever we find in that direction is even more exciting. (See Theorem 7.20 for an example.)

(b) Equivalence

Different sets of data "A" and "B" about a geometric situation are *equivalent* if any two configurations with the same data A also have the same data B, and conversely. This means that (at least in principle) one can construct the data A from the data B. Any set of data that is equivalent to the set of circuits is referred to as the *oriented matroid* of V.

We now want to show that the four sets of data for an oriented matroid given by Definition 6.5 are equivalent. For that we need to define the combinatorial analogue of the condition $\boldsymbol{cx} = 0$.

Definition 6.6. Let $\mathrm{X} \in \{+,-,0\}^n$ and $\mathrm{C} \in (\{+,-,0\}^n)^*$ be two sign vectors. Then we define that "$\mathrm{C}\cdot\mathrm{X} = 0$" if

- for each i, we have $c_i = 0$ or $x_i = 0$,
- or there are indices i, j with $c_i = x_i \neq 0$ and $c_j = -x_j \neq 0$.

For a family of sign vectors $\mathcal{S} \subseteq \{+,-,0\}^n$, we define

$$\mathcal{S}^\perp \;:=\; \{\mathrm{C} \in (\{+,-,0\}^n)^* : \mathrm{C}\cdot\mathrm{U} = 0 \text{ for all } \mathrm{U} \in \mathcal{S}\},$$

and analogously for collections of row vectors.